CENTRAL BOARD OF SECONDARY EDUCATION

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INFORMATION TECHNOLOGY (402)

PROJECT FILE

On

LIBRARY MANAGEMENT SYSTEM

LIBRE OFFICE BASE & WRITER

SUBMITTED BY :

S. No	Name	CBSE Roll
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CLASS : X
SECTION(A/B) :

INTERNAL EXAMINER
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(D.S RAWAT)

ACKNOWLEDGEMENT

*I wish to express my deep sense of gratitude and indebtedness to our learned teacher **Pankaj Bhilingwal**, PGT Computer Science, J.N.V Tarikhet Almora for his invaluable help, advice and guidance in the preparation of this project file.*

*I am also greatly indebted to our principal **D.S Rawat** and school authorities for providing me with the facilities and requisite laboratory conditions for making this project file.*

I also extend my thanks to a number of teachers, my classmates and friends who helped me to complete this project file successfully.

NAME :

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CERTIFICATE

This is to certify that,

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Student of Class X, J.N.V Tarikhet Almora has completed the PROJECT FILE during the academic year 2025-26 towards partial fulfilment of credit for the Information Technology Project evaluation of CBSE and submitted satisfactory report, as compiled in the following pages, under my supervision.

Internal Examiner
Signature

PREFACE

In the field of Computer Science, the term PROJECT refers to a Computerization Work that is completely performed by using a computer. Today when every field is going to be computerized means that all task of any field now a days done by using computer and their appropriate computer programs, then to do every field's different-different works, today different programs or software's are available in the form of Project.

Learning Libre Office in schools is a part of our syllabus, and according to the syllabus worked out by the CBSE, to which my school is affiliated, I enjoyed doing project work for the session 2025-26. I learned a lot, especially learning by doing was a great experience. The report is a summary of what I observed and learnt in school. My humble thanks are going to all the concerned related to my project.

I also feel that it is the right time to thank my teacher, who guided me during my project. I also feel that it is the right time to thank my parents, friends and my project partners for their constant assistance, cooperation extended by them time to time.

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About Project

Introduction:

The Library Management System is designed to efficiently manage the operations of a library, including the tracking of books, users, and transactions. This system facilitates the borrowing and returning of books, keeps records of users and books, and manages fines for overdue books.

Objective:

- Efficiently manage the library's book inventory.
- Facilitate user registration and management.
- Enable the borrowing and returning of books.
- Automate the calculation of fines for overdue books.
- Provide a user-friendly interface for library staff to interact with the system.

System Components:

The system consists of three main tables:

1. **Users Table:**
   - Fields: UserID (Primary Key), FirstName, LastName, Address, PhoneNumber, Email, MembershipStatus, MembershipExpiryDate.
2. **Books Table:**
   - Fields: BookID (Primary Key), Title, Author, ISBN, Genre, Pub Year, Qty, Shelf.
3. **Transactions Table:**
   - Fields: TransactionID (Primary Key), BookID (Foreign Key referencing Books Table), UserID (Foreign Key referencing Users Table), Transaction Type (Borrow or Return), Transaction Date, Due Date, Return Date, Fine Amount.

Database Design:

1. **Users Table:**
   - Holds information about library users.
   - UserID is the primary key, ensuring uniqueness.
   - MembershipExpiryDate tracks the expiration date of a user's membership.
2. **Books Table:**
   - Stores details about each book in the library.
   - BookID is the primary key.
   - Qty represents the quantity of each book available.
3. **Transactions Table:**
   - Manages the borrowing and returning of books.
   - TransactionID is the primary key.

- Foreign keys (BookID and UserID) establish relationships with the Books and Users tables.

System Functionality:

User Management:

- Users can be registered with details like name, address, phone number, and email.
- Membership status (Regular, Premium) and expiry date are tracked.

Book Management:

- Books can be added to the system with details like title, author, ISBN, genre, publication year, quantity, and shelf location.
- Availability and quantity are updated when books are borrowed or returned.

Transaction Handling:

- Transactions are recorded for each book borrowed or returned.
- Due dates are calculated based on the transaction date, and fines are applied for overdue books.
- Return dates and fine amounts are updated when books are returned.

System Implementation:

- The system is implemented using LibreOffice Base for database management.
- SQL commands are used to create tables, establish relationships, and insert data.
- Queries and forms can be created for easier data retrieval and entry.

Challenges Faced:

- Ensuring data integrity and consistency.
- Handling foreign key relationships between tables.
- Implementing an efficient fine calculation mechanism.

Future Enhancements:

- Implementing a reservation system for popular books.
- Incorporating a user authentication system for better security.
- Adding reporting features for library administrators.

Conclusion:

The Library Management System efficiently manages the library's operations, providing an organized and user-friendly interface for both library staff and users. It automates routine tasks, enhances data accuracy, and contributes to the overall efficiency of the library.

Table : Relationship

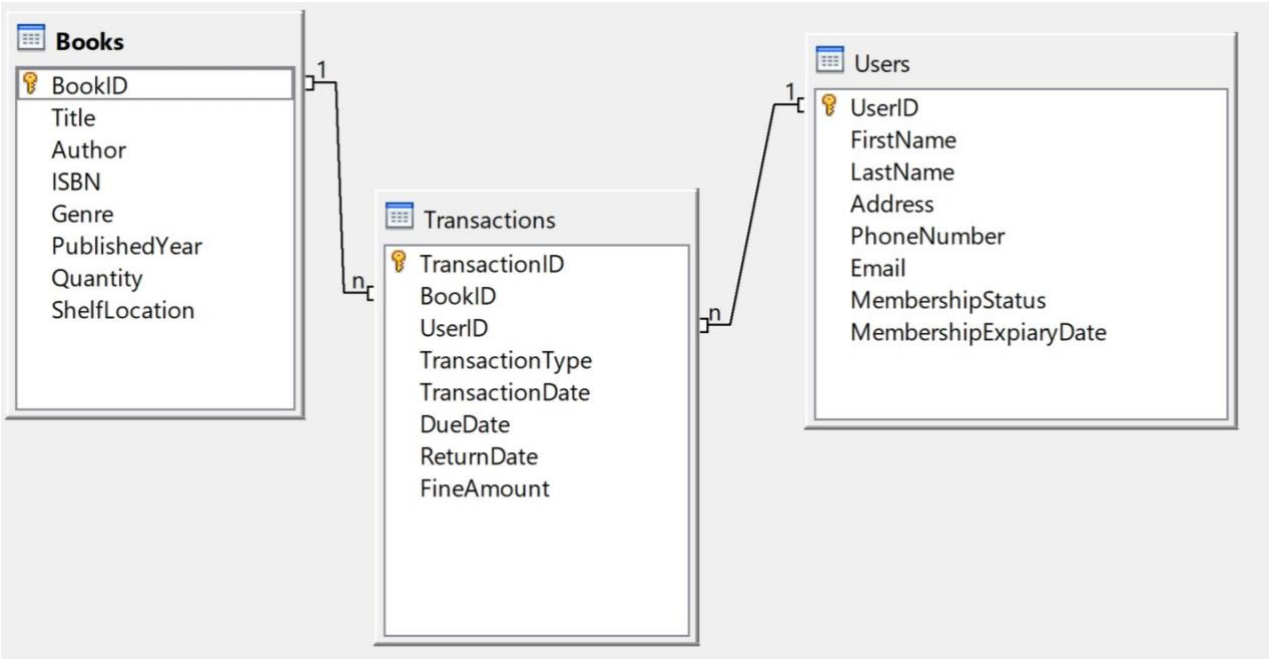
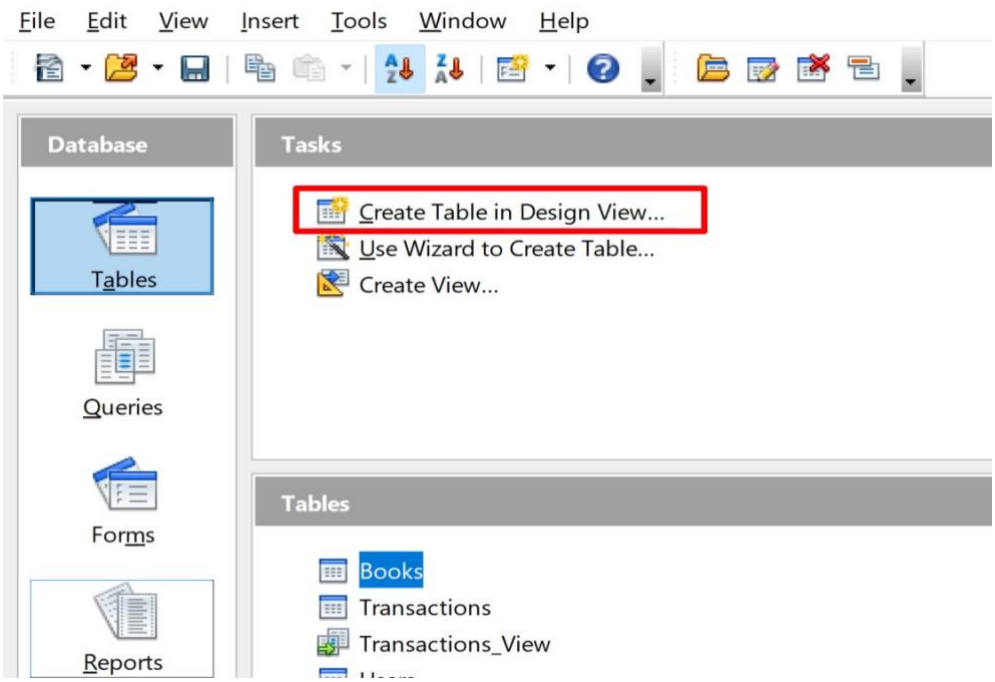


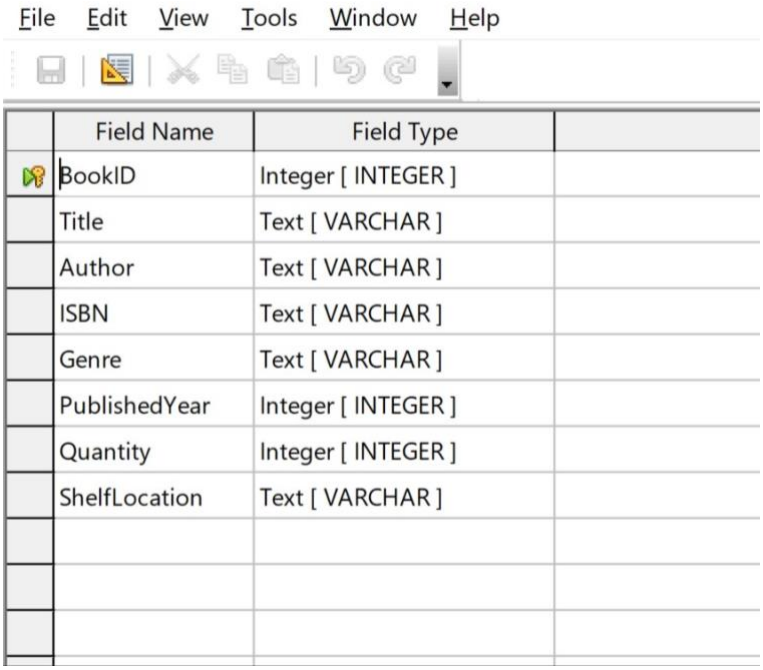
Table Creation Steps

Creating a table in LibreOffice Base involves a few steps. LibreOffice Base typically uses the HSQLDB (Hypersonic SQL Database) engine, and you can create tables through the graphical user interface. Here are the steps to create a table in LibreOffice Base:

1. **Open LibreOffice Base:**
    - Launch LibreOffice Base.
2. **Connect to a Database:**
    - Open an existing database or create a new one.
    - Go to "File" > "New" > "Database..."
3. **Create a New Table:**
    - In the Database Wizard, choose "Create a new database" or select an existing one.
    - Click "Finish."
4. **Switch to Tables View:**
    - In the main LibreOffice Base window, go to the "Tables" section.
5. **Create a Table:**
    - Right-click in the "Tables" section and select "Create Table."
    -
    - This will open the "Table Design" window.



6. **Define Fields:**
    - In the "Table Design" window, define the fields of your table.
    - Enter the field name in the "Field Name" column.
    - Choose the data type for each field from the "Type" column (e.g., INT, VARCHAR, DATE, etc.).
    - You can also set additional properties like primary key, not null, etc.



7. **Set Primary Key:**
    - Right-click on a field and select "Primary Key" if applicable.
8. **Save the Table:**
    - Click the "Save" icon or go to "File" > "Save Table Design."
    - Enter a name for your table and click "OK."
9. **Add Data:**
    - After saving the table structure, switch to the "Data View."
    - Here, you can enter data directly into the table.

BookID	Title	Author	ISBN	Genre	PublishedYear	Quantity	ShelfLocation
1	The Great Gatsby	F. Scott Fitzgerald	978-3-16-148410-0	Fiction	1925	5	A1
2	To Kill a Mockingbird	Harper Lee	978-0-06-112008-4	Fiction	1960	3	B2
3	1984	George Orwell	978-0-45-152493-5	Science Fiction	1949	7	C3
4	Pride and Prejudice	Jane Austen	978-0-14-143951-8	Romance	1813	4	A2
5	The Catcher in the Rye	J.D. Salinger	978-0-31-676948-0	Fiction	1951	6	B1

Book Table

UserID	FirstName	LastName	Address	PhoneNumber	Email	MembershipStatus	MembershipExpiaryDate
1	John	Doe	123 Main St	123-456-7890	john.doe@email.com	Regular	31/12/23
2	Jane	Smith	456 Oak St	987-654-3210	jane.smith@email.com	Premium	15/03/24
3	Bob	Johnson	789 Pine St	555-123-4567	bob.johnson@email.com	Regular	22/10/23

User Table

TransactionID	BookID	UserID	TransactionType	TransactionDate	DueDate	ReturnDate	FineAmount
1	1	1	Borrow	01/01/23	15/01/23		
2	2	2	Borrow	01/02/23	15/02/23		
3	3	3	Borrow	01/03/23	15/03/23	10/03/23	5
4	4	1	Return	15/04/23	15/04/23	14/04/23	1.5
5	5	2	Return	15/05/23	15/05/23		
6	1	3	Borrow	01/06/23	15/06/23		
7	2	3	Borrow	01/07/23	15/07/23		
8	5	2	Borrow	01/08/23	15/08/23	10/08/23	7.5
9	4	1	Return	15/09/23	15/09/23	14/09/23	2
10	5	2	Return	15/10/23	15/10/23		
0							

Trasaction Table

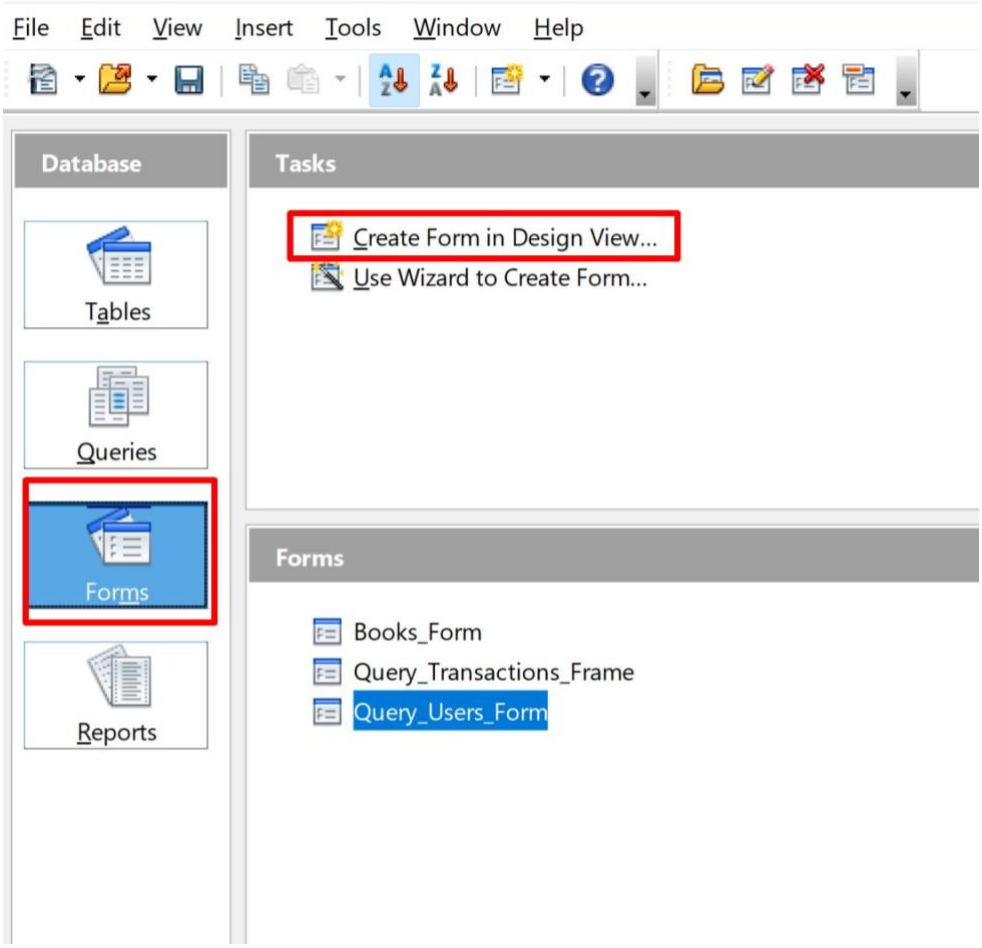
10. **Save and Close:**

- Save your changes by clicking the "Save" icon or going to "File" > "Save."
- Close the table design window.

Form Creation Step

Creating a form in LibreOffice Base allows users to interact with the data in a more user-friendly manner. Here are the steps to create a form in LibreOffice Base:

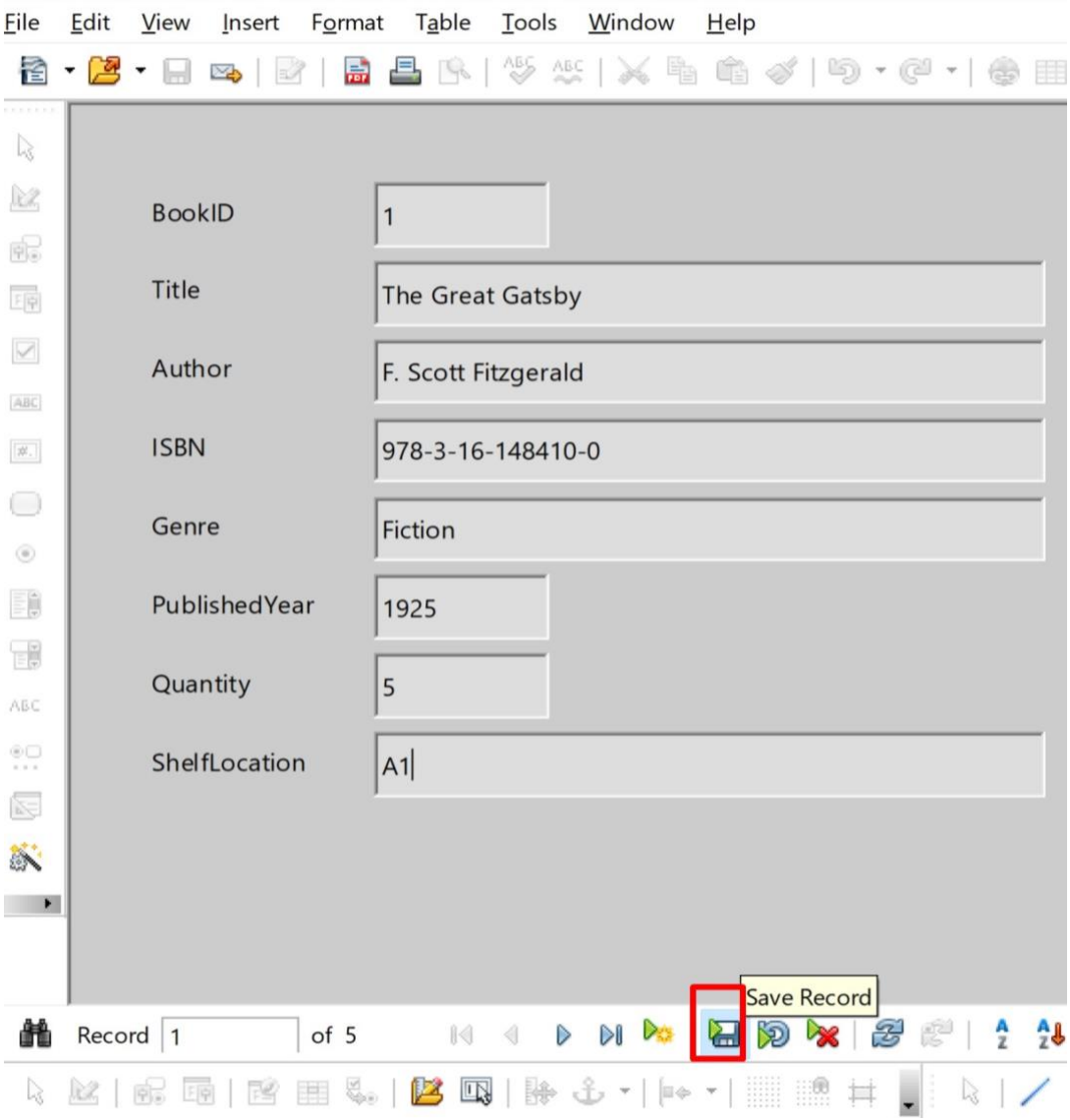
1. **Open LibreOffice Base:**
    - Launch LibreOffice Base.
2. **Navigate to Forms:**
    - In the main LibreOffice Base window, go to the "Forms" section.
3. **Create a New Form:**
    - Right-click in the "Forms" section and select "Use Wizard to Create Form..."
    - This will open the Form Wizard.



4. **Select a Table or Query:**
    - In the Form Wizard, choose the table or query that you want the form to be based on.
    - Click "Next."
5. **Select Fields:**

- Choose the fields from the selected table or query that you want to include in the form.
- Use the arrow buttons to move fields between the "Available Fields" and "Fields in the Form" boxes.
- Click "Next."

6. **Set Form Style:**
    - Choose a style for your form (e.g., Columnar, Tabular).
    - Click "Next."
7. **Set Subforms (Optional):**
    - If your data includes related tables, you can include subforms.
    - Click "Next."
8. **Define Sorting and Grouping (Optional):**
    - You can define sorting and grouping options for your form.
    - Click "Next."
9. **Choose Form Title:**
    - Enter a title for your form.
    - Click "Finish."
10. **Modify Form Layout (Optional):**
    - After the Form Wizard completes, you can modify the form layout by dragging and dropping fields.
    - Right-click on the form and select "Edit" to make adjustments.
11. **Save and Close:**
    - Save your form by clicking the "Save" icon or going to "File" > "Save."
    - Close the form design window.
12. **Test the Form:**
    - Switch to the "Forms" section in LibreOffice Base.
    - Double-click on your form to open it and test its functionality.

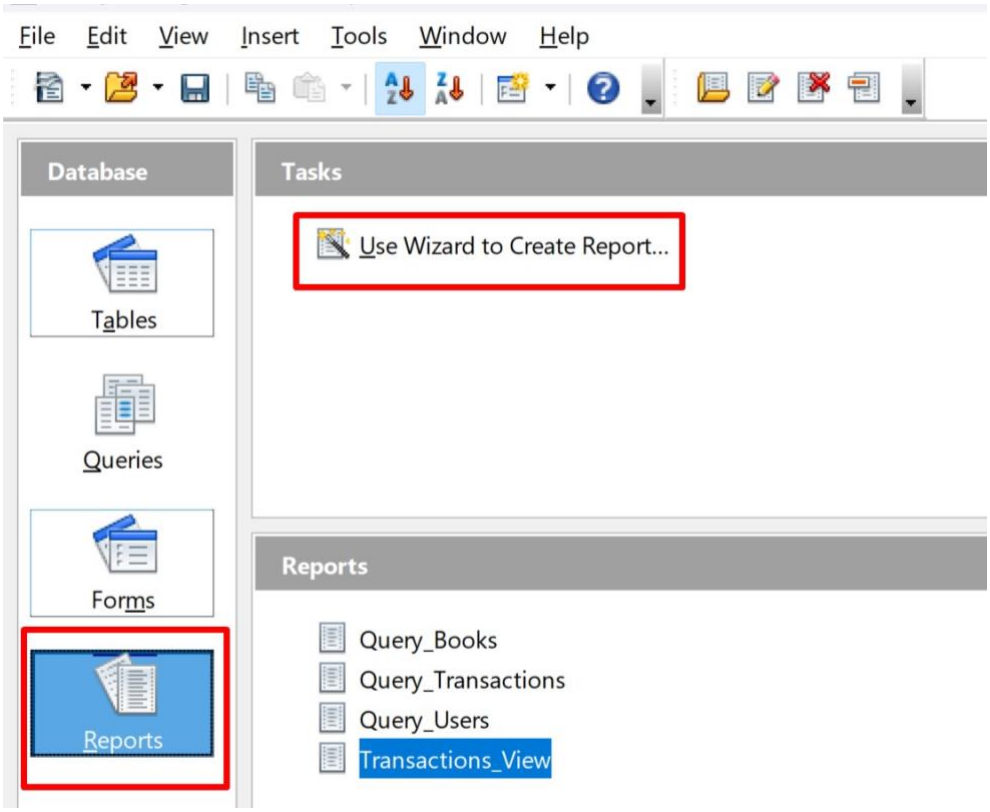


13. **Further Customization (Optional):**
    - You can further customize the form by adding buttons, labels, and adjusting properties.
    - Explore the form controls in the toolbar to add functionality.
14. **Save Database:**
    - Save your changes to the database by clicking "File" > "Save All."

Report Generation Steps

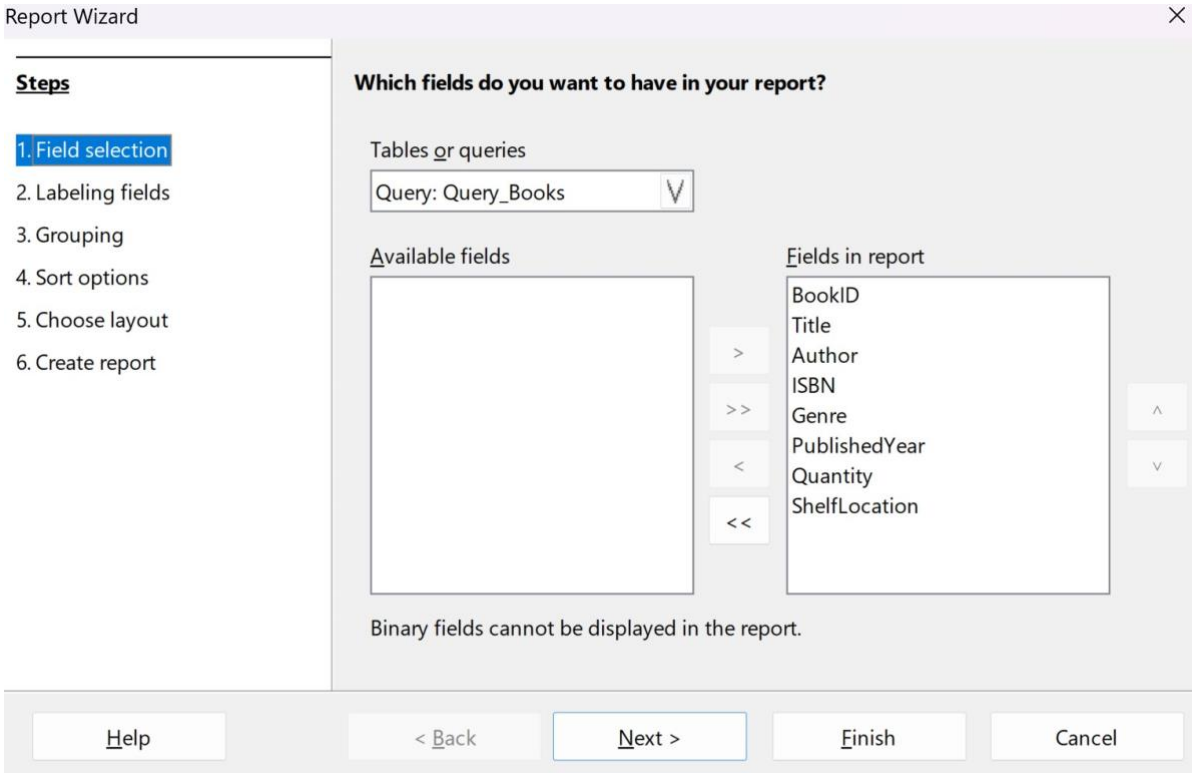
Creating a report in LibreOffice Base allows you to present and analyse data in a structured format. Here are the steps to create a report in LibreOffice Base:

1. **Open LibreOffice Base:**
    - Launch LibreOffice Base.
2. **Navigate to Reports:**
    - In the main LibreOffice Base window, go to the "Reports" section.



3. **Create a New Report:**
    - Right-click in the "Reports" section and select "Use Wizard to Create Report..."
    - This will open the Report Wizard.
4. **Select a Table or Query:**
    - In the Report Wizard, choose the table or query that you want the report to be based on.
    - Click "Next."

5. **Select Fields:**
    - Choose the fields from the selected table or query that you want to include in the report.
    - Use the arrow buttons to move fields between the "Available Fields" and "Fields in the Report" boxes.
    - Click "Next."
6. **Define Grouping and Sorting (Optional):**
    - You can define grouping and sorting options for your report.
    - Click "Next."



7. **Choose Report Layout:**
    - Choose a layout for your report (e.g., Columnar, Tabular, Grouped).
    - Click "Next."
8. **Set Column Widths (Optional):**
    - If you selected a tabular layout, you can set column widths.
    - Click "Next."
9. **Set Subreports (Optional):**
    - If your data includes related tables, you can include subreports.

- Click "Next."

10. **Define Calculations (Optional):**
    - You can define calculations for fields in your report.
    - Click "Next."
11. **Choose Report Title:**
    - Enter a title for your report.
    - Click "Finish."
12. **Modify Report Layout (Optional):**
    - After the Report Wizard completes, you can modify the report layout by dragging and dropping fields.
    - Right-click on the report and select "Edit" to make adjustments.
13. **Save and Close:**
    - Save your report by clicking the "Save" icon or going to "File" > "Save."
    - Close the report design window.
14. **Test the Report:**
    - Switch to the "Reports" section in LibreOffice Base.
    - Double-click on your report to open it and test its functionality.
15. **Further Customization (Optional):**
    - You can further customize the report by adding labels, adjusting formatting, and applying styles.
16. **Save Database:**
    - Save your changes to the database by clicking "File" > "Save All."

Final Report

Book Report

BookID	1					
Title	***Author***	***ISBN***	***Genre***	***PublishedYear***	***Quantity***	***ShelfLocation***
The Great Gatsby	F. Scott Fitzgerald	0	Fiction	1925	5	A1
BookID	2					
Title	***Author***	***ISBN***	***Genre***	***PublishedYear***	***Quantity***	***ShelfLocation***
To Kill a Mockingbird	Harper Lee	0	Fiction	1960	3	B2
BookID	3					
Title	***Author***	***ISBN***	***Genre***	***PublishedYear***	***Quantity***	***ShelfLocation***
1984	George Orwell	0	Science Fiction	1949	7	C3
BookID	4					
Title	***Author***	***ISBN***	***Genre***	***PublishedYear***	***Quantity***	***ShelfLocation***
Pride and Prejudice	Jane Austen	0	Romance	1813	4	A2
BookID	5					
Title	***Author***	***ISBN***	***Genre***	***PublishedYear***	***Quantity***	***ShelfLocation***
The Catcher in the Rye	J.D. Salinger	0	Fiction	1951	6	B1

User Report

UserID	1					
FirstName	***LastName***	***Address***	***PhoneNumber***	***Email***	***MembershipStatus***	***MembershipExpiaryDate***
0	0	0	0	0	0	12/31/23
UserID	2					
FirstName	***LastName***	***Address***	***PhoneNumber***	***Email***	***MembershipStatus***	***MembershipExpiaryDate***
0	0	0	0	0	0	03/15/24

UserID	3					
FirstName	LastName	Address	PhoneNumber	Email	MembershipStatus	MembershipExpiaryDate
0	0	0	0	0	0	10/22/23

Transaction Report

UserID	1					
BookID	TransactionID	TransactionType	TransactionDate	DueDate	ReturnDate	FineAmount
1	1	Borrow	01/01/23	01/15/23		
4	9	Return	09/15/23	09/15/23	09/14/23	2
4	4	Return	04/15/23	04/15/23	04/14/23	1.5
UserID	2					
BookID	TransactionID	TransactionType	TransactionDate	DueDate	ReturnDate	FineAmount
2	2	Borrow	02/01/23	02/15/23		
5	10	Return	10/15/23	10/15/23		
5	8	Borrow	08/01/23	08/15/23	08/10/23	7.5
5	5	Return	05/15/23	05/15/23		
UserID	3					
BookID	TransactionID	TransactionType	TransactionDate	DueDate	ReturnDate	FineAmount
1	6	Borrow	06/01/23	06/15/23		
2	7	Borrow	07/01/23	07/15/23		
3	3	Borrow	03/01/23	03/15/23	03/10/23	5

Technological requirements: Hardware & Software

Software

1. Libre Office
2. Java Runtime Environment
3. Lighshot (for capturing screenshot)

Hardware

1. Laptop/PC

References

1. https://www.libreoffice.org/
2. https://www.youtube.com/watch?v=VTrcboJWzVM
3. https://www.youtube.com/watch?v=dyUrXvkQPWM&list=PLyH-qXFkNSxmEiBlIxdzc5n9DXnBc8_mX